

Implied Volatility

Implied volatility is determined from the price of a call or put option. For example, the Black-Scholes option pricing model determines the price of a call option as follows:

$$C = S \cdot N(d_1) - X \cdot e^{-r_f T} \cdot N(d_2) \quad (6.11)$$

where,

$$d_1 = \frac{\ln(S/X) + (r_f + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

C = call price

X = strike price

S = stock price

σ = stock volatility

$N(d)$ = probability that actual return will be less than d

r_f = risk free rate of return

T = future time period

The implied volatility is the value of the volatility in the above formula that will result in the current value of the call option. Since the call option price is determined by the market, we are able to back into the volatility terms that would provide this value, thus, the volatility is implied by the formulation. Implied volatility is most often solved via non-linear optimization techniques.

Beta

$$\beta_k = \frac{\text{cov}(r_k, r_m)}{\text{var}(r_m)} = \frac{\sigma_{kj}}{\sigma_m^2} \quad (6.12)$$

The beta of a stock represents the stock's sensitivity to a general market index. It is determined as the covariance of returns between the stock and the market divided by the variance of the index (volatility squared). The calculation is also the slope of the regression line of stock returns (y-axis) as a function of market returns (x-axis). Stocks with a positive beta, $\beta_k > 0$, move in the same direction as the market and stocks with a negative beta, $\beta_k < 0$, will move in the opposite direction of the market. Stocks with an absolute value of beta greater than one, $|\beta_k| > 1$, are more variable than the market and stocks with an absolute value of beta less than one, $|\beta_k| < 1$, are less variable than the market.